Advanced Stock Market Analysis Dashboard

Using Python & Streamlit

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1. Project Overview

This project is an advanced web-based application developed using Streamlit for analyzing stock market data. The system provides interactive visualizations, real-time insights, and a modern dashboard interface that enhances the user experience and understanding of stock performance..

2. Objectives

- To understand how stock market data is retrieved and processed from APIs
- To build an interactive and user-friendly financial dashboard
- To visualize stock price trends using advanced charts (such as candlestick charts)
- To provide meaningful insights including price changes and comparisons

3. Scope

The system allows users to:
- Search for stock symbols (e.g., AAPL, TSLA)
- Select different time periods
- View real-time and historical stock data
- Analyze trends using charts
- Compare multiple stocks

4. Functional Requirements

The application should:
- Accept stock symbol input from the user
- Retrieve stock data using Yahoo Finance API (via yfinance)
- Display current stock price and key metrics
- Show historical data for selected periods
- Generate interactive charts (line and candlestick)
- Provide optional features like moving average and stock comparison
- Handle invalid inputs with proper error messages

5. Non-Functional Requirements

- The system should have a modern and responsive design
- Data should load quickly and efficiently
- The system should be reliable and stable
- The interface should be clear and easy to use

6. Tools & Technologies

The project was developed using modern Python-based technologies to create an interactive and responsive stock analysis dashboard.
The main development framework used was Streamlit, which simplified the process of building the web interface and connecting backend logic with data visualization components.

Several supporting libraries were integrated into the system:

- Pandas was used for cleaning, organizing, and processing stock market datasets.
- Plotly was implemented to generate interactive financial charts such as candlestick charts and comparison graphs.

- yfinance was used as the primary source for retrieving real-time and historical stock market data from Yahoo Finance.
- Python served as the core programming language responsible for data processing and application logic.

These technologies together allowed the system to deliver a smooth user experience with fast and accurate financial analysis.

7.System Workflow

The system follows a simple but efficient workflow to process stock market information and display analytical results to the user.

1. The user enters a stock symbol such as TSLA or AAPL through the dashboard interface.
2. The application validates the entered symbol to ensure it is supported.
3. The system sends a request to the Yahoo Finance API to retrieve market data.
4. The retrieved data is processed and filtered using Pandas.
5. Financial metrics such as price changes, highs, lows, and moving averages are calculated.
6. Interactive visualizations and charts are generated dynamically using Plotly.
7. The final output is displayed through the Streamlit dashboard in a clean and user-friendly format.

This workflow enables users to analyze stock performance quickly and efficiently.

8.User Interface Requirements

The user interface was designed to provide a modern and intuitive experience while keeping financial data easy to understand.

The dashboard includes:

- A sidebar section for entering stock symbols and selecting time periods.
- KPI cards displaying important values such as current price, percentage change, and period high.
- Interactive candlestick charts with optional moving average overlays.
- A stock comparison section allowing users to compare multiple companies visually.
- Expandable panels for raw data tables and company information.
- A responsive dark-themed layout optimized for readability and professional appearance.

The interface prioritizes simplicity, responsiveness, and accessibility for all users.

9.Testing Requirements

Several testing procedures were performed to ensure the reliability and accuracy of the application.

The testing process included:

- Verifying that valid stock symbols return correct market data.

- Testing invalid or unsupported symbols to ensure proper error handling.
- Checking the accuracy of generated charts and financial metrics.
- Evaluating dashboard responsiveness under different user interactions.
- Testing data loading speed and overall application performance.
- Ensuring compatibility across different screen sizes and browsers.

These tests helped improve system stability and user experience.

10. Expected Output

The application is expected to provide users with a complete and interactive stock market analysis experience.

The final output includes:

- Real-time stock price information and market statistics.
- Interactive candlestick and comparison charts.
- Visual representation of stock trends over selected periods.
- Clear and responsive dashboard components.
- Organized financial data presented in an easy-to-read format.
- Meaningful insights that help users understand stock performance and market behavior.

6. Conclusion

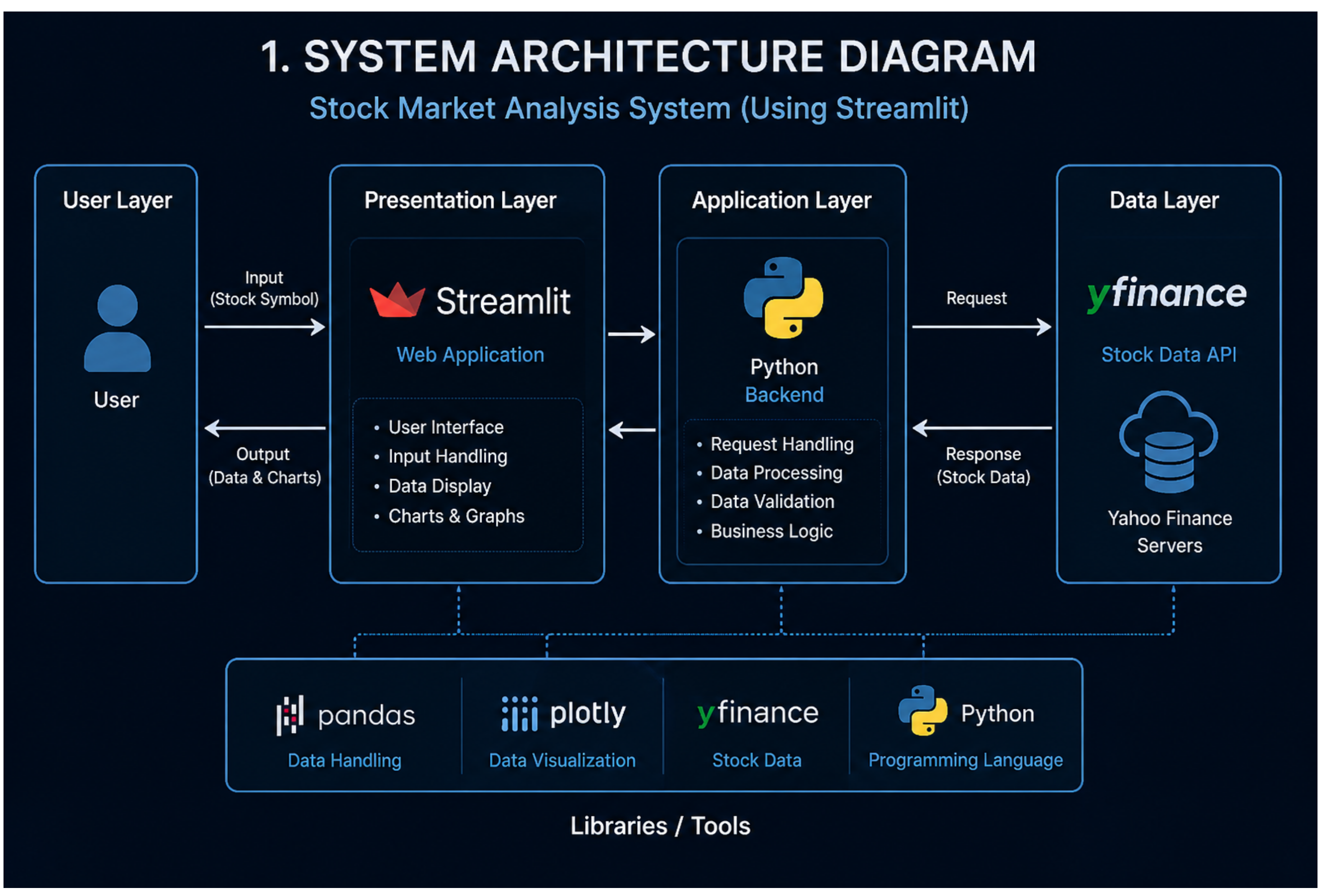
This project successfully demonstrates the development of a modern stock market analysis dashboard using Python and Streamlit.
The system combines real-time financial data, interactive visualizations, and a professional user interface to provide users with an effective way to analyze stock market performance.

Through the integration of technologies such as Pandas, Plotly, and yfinance, the application is capable of processing and presenting financial information in a clear and user-friendly manner.
Features such as candlestick charts, stock comparison, KPI metrics, and responsive dashboard components enhance both usability and analytical capabilities.

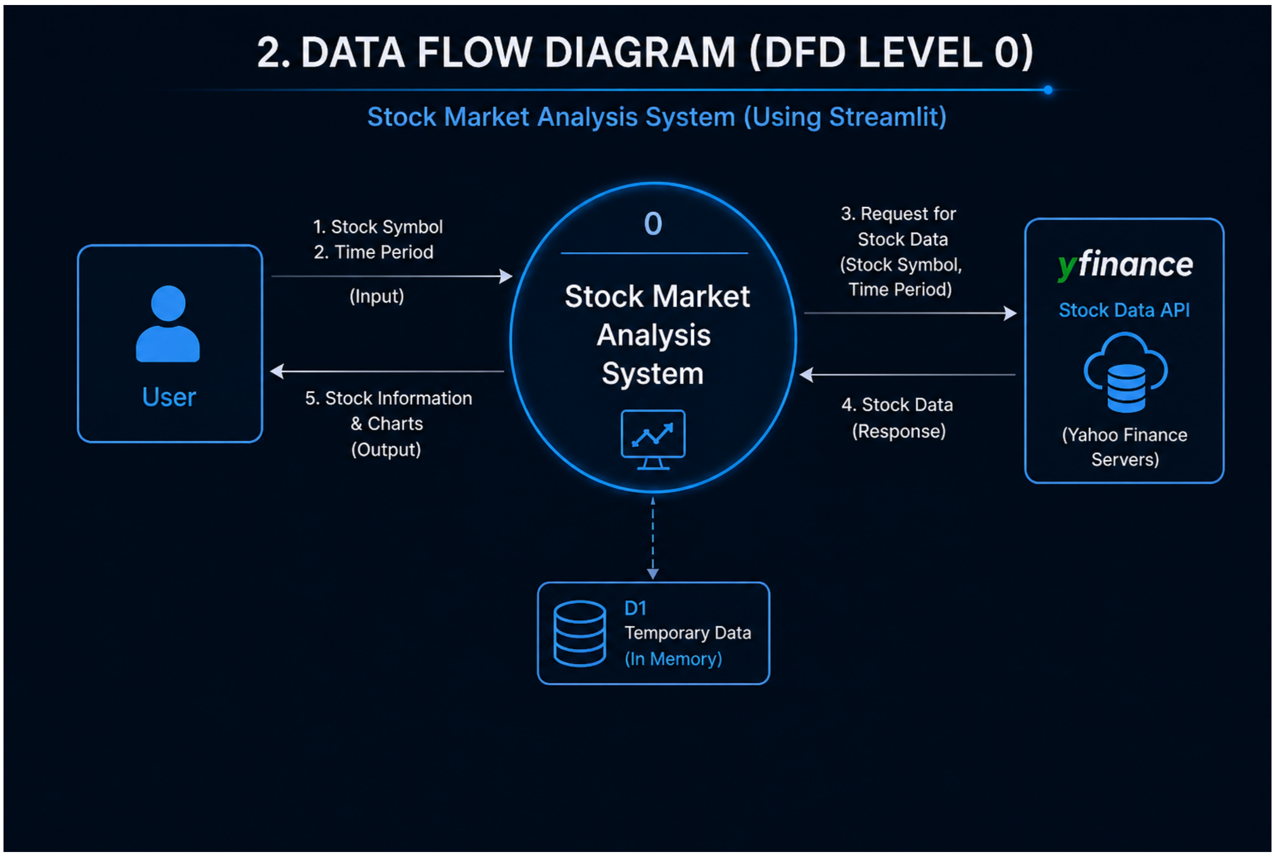
7. Diagrams Included

The following diagrams were designed:

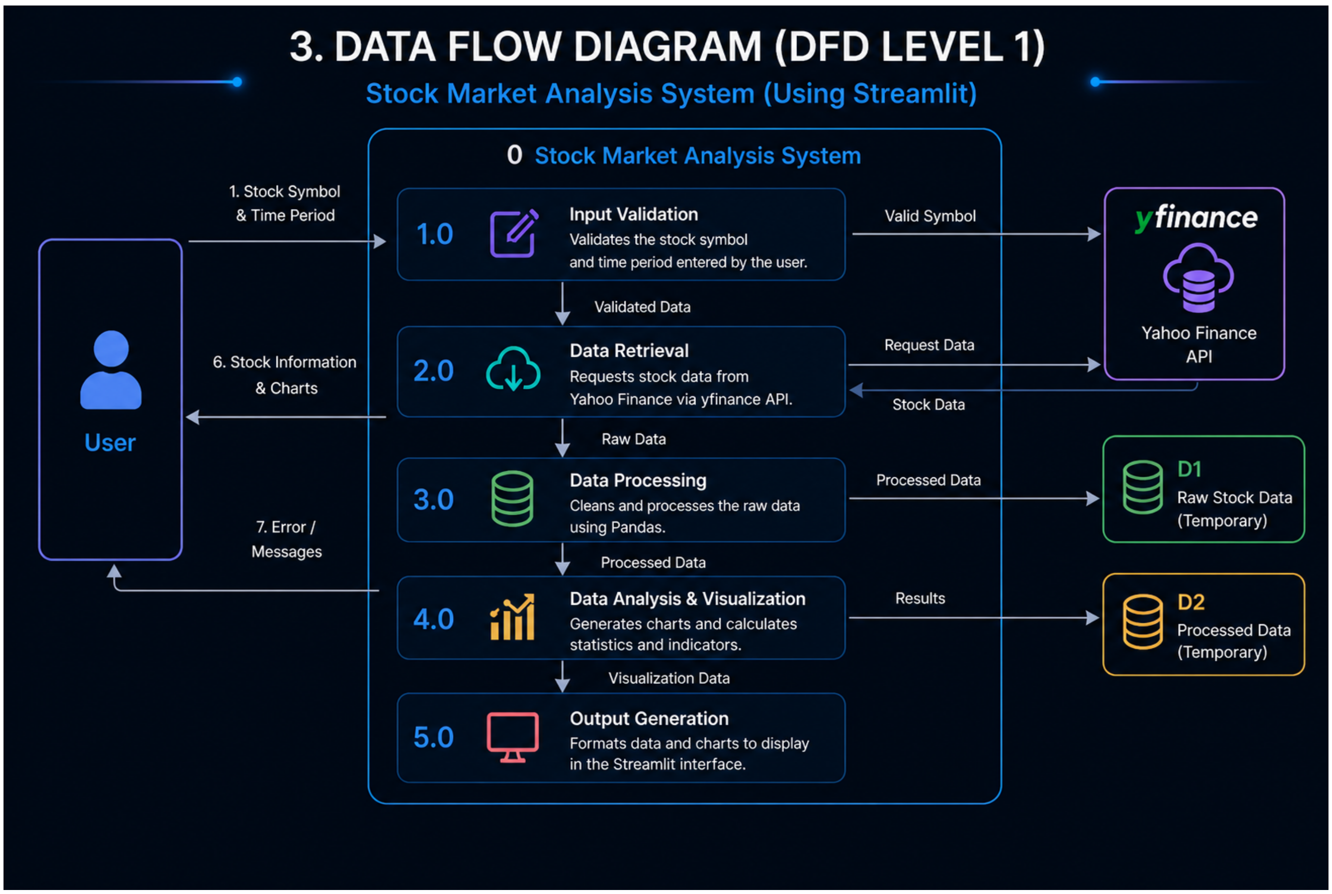
System Architecture Diagram



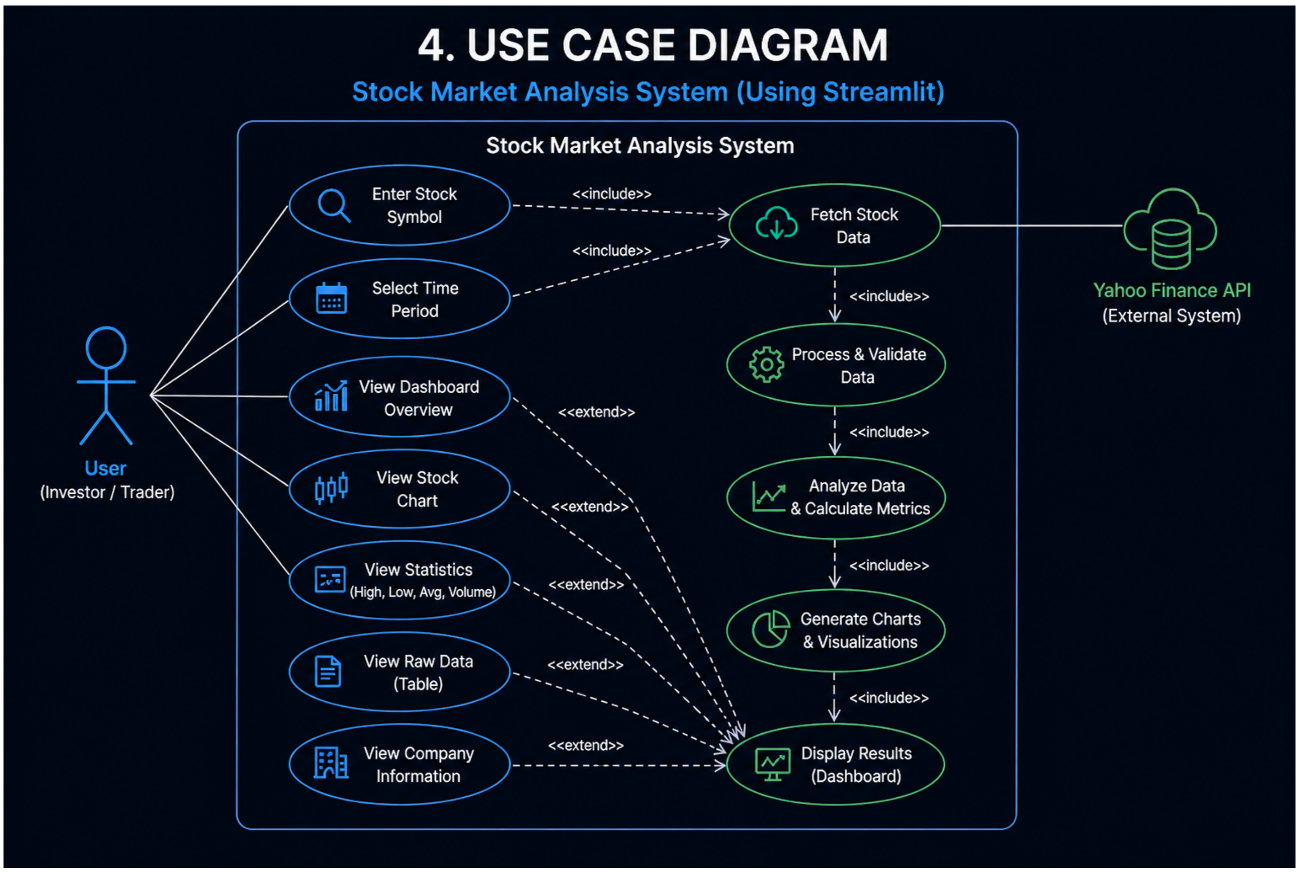
Data Flow Diagram (Level 0)



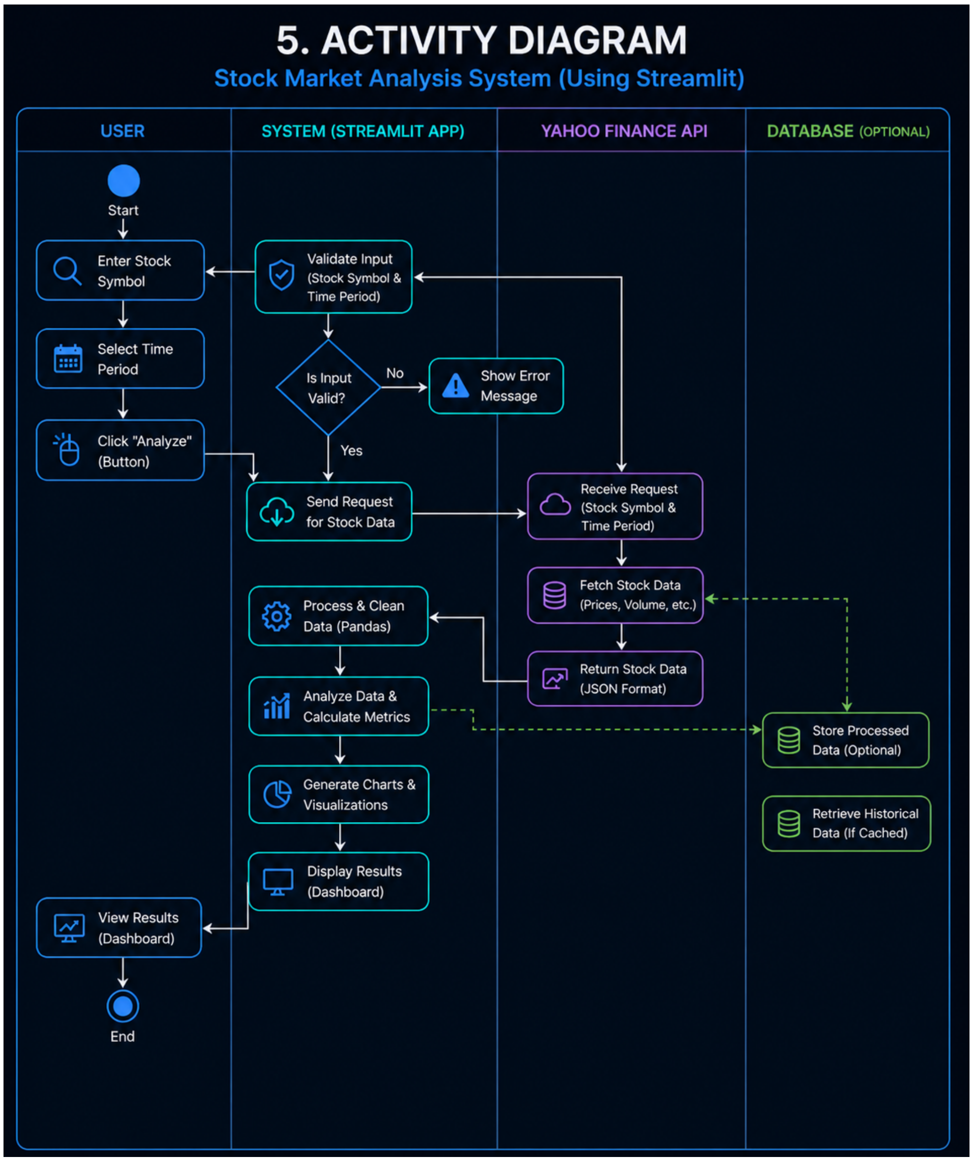
Data Flow Diagram (Level 1)



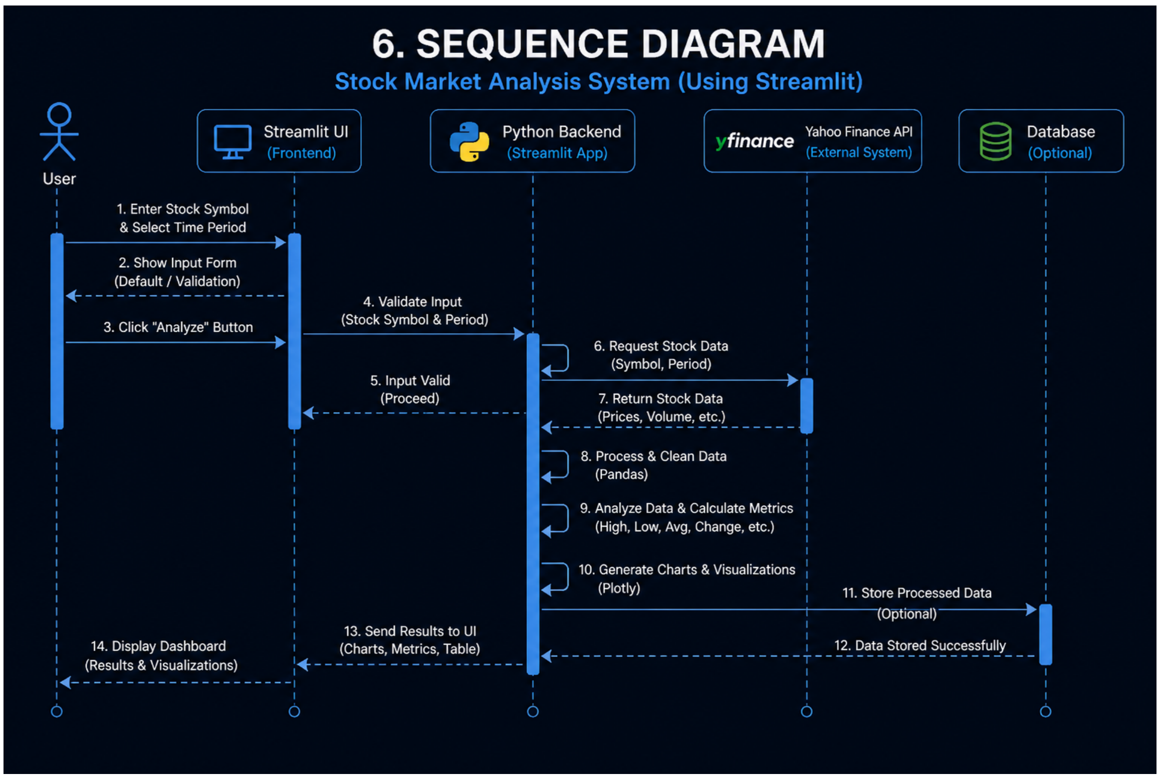
Use Case Diagram



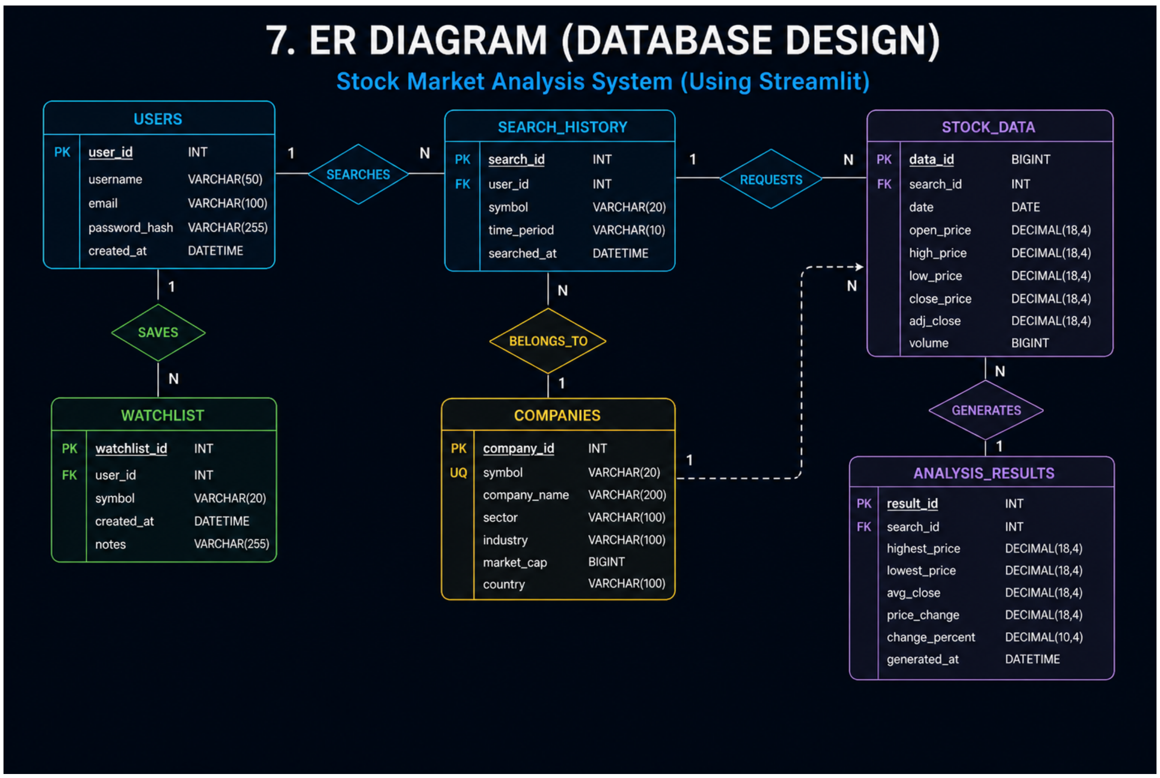
Activity Diagram



Sequence Diagram

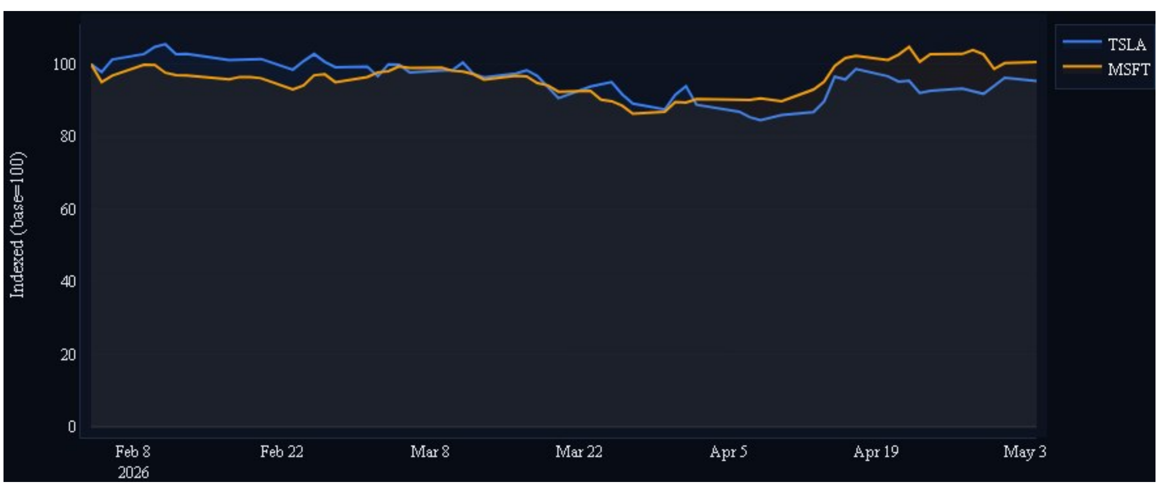


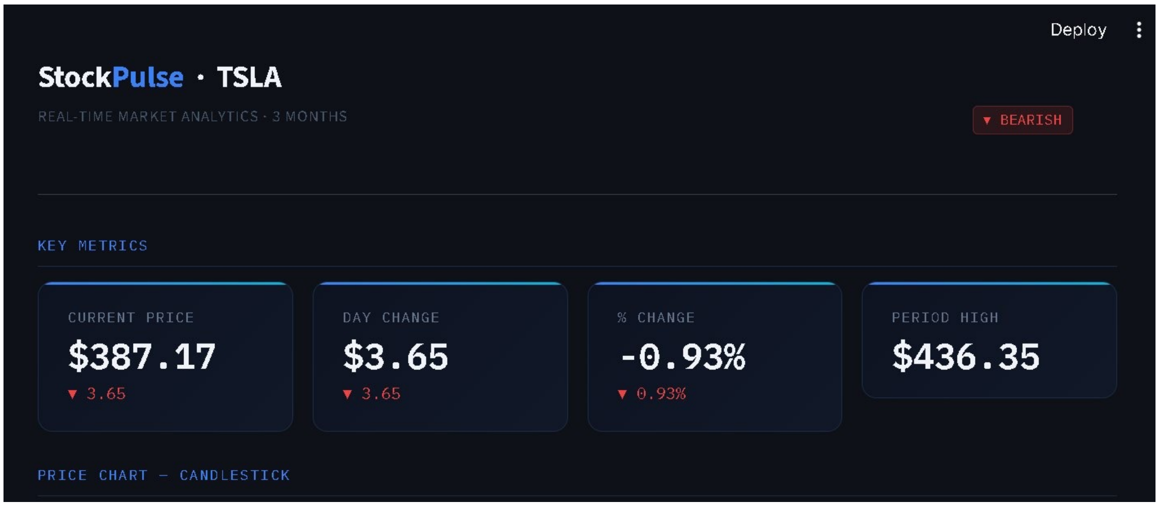
ER Diagram



7. ER DIAGRAM (DATABASE DESIGN)
Stock Market Analysis System (Using Streamlit)
USERS
PK user_id INT
username VARCHAR(50)
email VARCHAR(100)
password_hash VARCHAR(255)
created_at DATETIME
SEARCHES
1
N
SEARCH_HISTORY
PK search_id INT
FK user_id INT
symbol VARCHAR(20)
time_period VARCHAR(10)
searched_at DATETIME
REQUESTS
1
N
STOCK_DATA
PK data_id BIGINT
FK search_id INT
date DATE
open_price DECIMAL(18,4)
high_price DECIMAL(18,4)
low_price DECIMAL(18,4)
close_price DECIMAL(18,4)
adj_close DECIMAL(18,4)
volume BIGINT
SAVES
1
N
WATCHLIST
PK watchlist_id INT
FK user_id INT
symbol VARCHAR(20)
created_at DATETIME
notes VARCHAR(255)
BELONGS_TO
N
1
COMPANIES
PK company_id INT
UQ symbol VARCHAR(20)
company_name VARCHAR(200)
sector VARCHAR(100)
industry VARCHAR(100)
market_cap BIGINT
country VARCHAR(100)
1
N
GENERATES
N
1
ANALYSIS_RESULTS
PK result_id INT
FK search_id INT
highest_price DECIMAL(18,4)
lowest_price DECIMAL(18,4)
avg_close DECIMAL(18,4)
price_change DECIMAL(18,4)
change_percent DECIMAL(10,4)
generated_at DATETIME

8. Screenshots




Deploy
StockPulse · TSLA
REAL-TIME MARKET ANALYTICS · 3 MONTHS
▼ BEARISH
KEY METRICS
CURRENT PRICE
$387.17
▼ 3.65
DAY CHANGE
$3.65
▼ 3.65
% CHANGE
-0.93%
▼ 0.93%
PERIOD HIGH
$436.35
PRICE CHART — CANDLESTICK